

Homework #3

MH sampling: joint moves versus full conditioning

1. Use the MH-logit codes we used in class (IllustrateRWMH.R and rmnlRWMetrop_noTuning.r) as a basis for an MH-sampler for the MNL model that generates from the *full conditional distributions of elements in the parameter vector beta* (in a loop over the elements of the parameter vector). Remember that conditional distributions are *proportional* to the joint distribution. Experiment with different step sizes and compare the acceptance rates and autocorrelations of the original “multi move” MH and your “single move” MH sampler.
2. Change the prior in this model from a multivariate normal prior to a truncated multivariate normal prior constraining the first, second, and third element of the parameter vector to be negative, positive, and positive, respectively (in line with the data generating values $\beta = c(-0.5, 0.5, 0.2)$ in the script IllustrateRWMH.R). Convince yourself analytically that MH-sampling does not require the normalizing constant of the truncated multivariate normal prior. As a consequence, you can again use a “multi move” or a “single move” MH sampler. i) Compare acceptance rates and autocorrelations between the “multi move” and the “single move” MH sampler for the MNL model with a truncated multivariate normal prior. ii) Compare the posterior distribution and the model fit from the model with the truncated multivariate normal prior to what you got in question 1.
3. Finally, use the sampler(s) and the model developed in question 2, but use constraints that are inconsistent with the data generating parameters, i.e., constrain the first, second, and third element of the parameter vector to be positive, negative, and negative, respectively. Compare the posterior distribution and the model fit from this specification to what you got in questions 1 and 2.